

Locally Equivalent Weights for Multilevel Regression and Poststratification

Ryan Giordano, Alice Cima, Erin Hartman, Jared Murray, Avi Feller

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Running example: Same-Sex Marriage Survey

Analysis of US state-level support for same-sex marriage¹.

- **Survey population:** Five consistently-coded national polls from 2004 ($N_S = 6,341$)
- **Response:** $y = 1$ encodes support for same-sex marriage, $y = 0$ opposition or no opinion.
- **Target population:** US Census 5% PUMS data for CA from 2000

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MrP computed with `brms` [Bür17]:

$$y \sim \text{logit}\left((1 \mid \text{race.female}) + (1 \mid \text{age.cat}) + (1 \mid \text{edu.cat}) + (1 \mid \text{age.edu.cat}) + \right. \\ \left. (1 \mid \text{state}) + (1 \mid \text{region}) + (1 \mid \text{poll}) + \text{p.relig.full} + \text{p.kerry.full} \right)$$

CW used raking on the following coarsened regressors: (`survey::calibrate` Lumley [Lum24]):

`female, edu.cat, age.cat, race.wbh, race.w.female, age.edu.low`

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Raw survey mean: $\frac{1}{N_S} \sum_{i=1}^{N_S} y_i = 0.333$

Raking: $\hat{\mu}^{\text{CW}}(Y_S) = 0.345$

MrP: $\hat{\mu}^{\text{MrP}}(Y_S) = 0.457$

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How are MrP and raking using the data differently?

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Prior work: Equivalent weights for linear models

Observe $(\mathbf{x}_i, y_i)$ for $i = 1, \dots, N_S$

Let $Y_S = \{y_1, \dots, y_{N_S}\}$.

Observe $\mathbf{x}_j$ for $j = 1, \dots, N_T$.

We want $\mu := \frac{1}{N_T} \sum_{j=1}^{N_T} y_j$, but don't observe y_j .

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- Frequentist variability
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When using OLS, $\hat{y}_j(Y_S)$ is linear [Gel07b; BFH21; CZ23], so MrP is a CW estimator.

But currently, for (ubiquitous) MrP models like hierarchical logistic regression, no such equivalent weights are available.[Gel07a]

Linear and nonlinear models

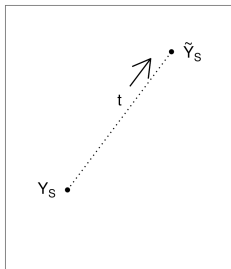
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Path through response space



Linear and nonlinear models

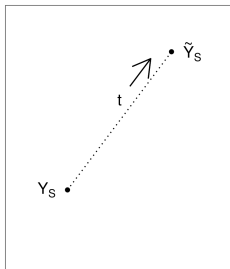
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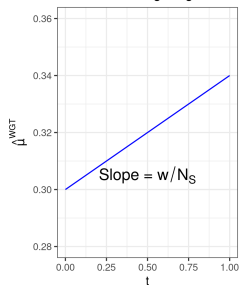
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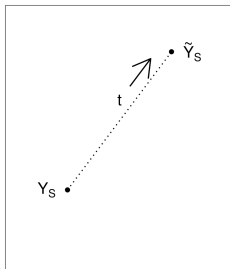
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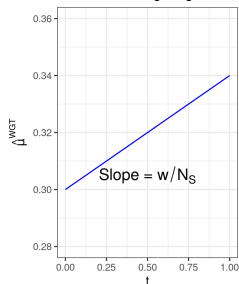
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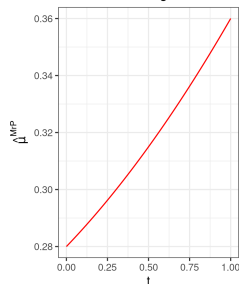
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Calibration weighting



Hierarchical / logistic MrP



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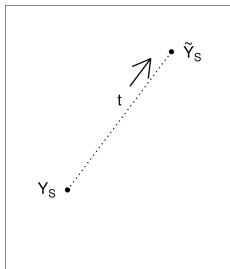
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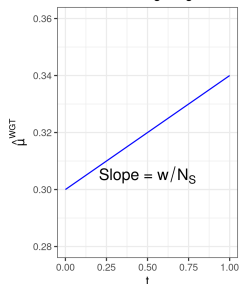
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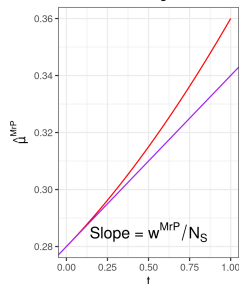
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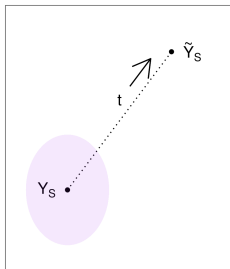
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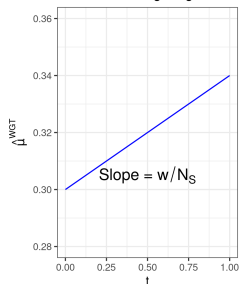
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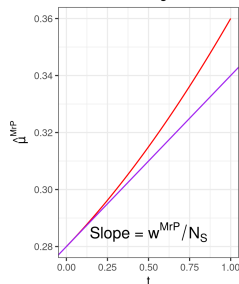
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The slope is a *MrP locally equivalent weight*, or *MrPlew*.

Linear and nonlinear models

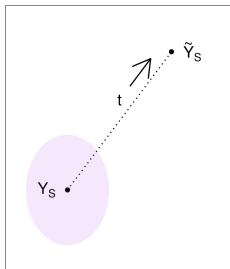
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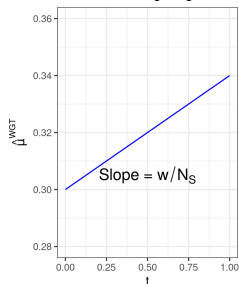
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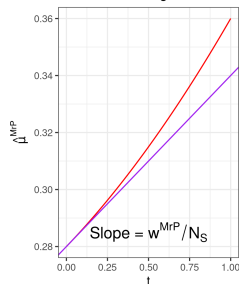
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The weight w_i^{MrP} measures the effect of local perturbations to y_i on the estimator $\hat{\mu}^{MrP}$.

Our main contribution is to rigorously connect this idea to CW diagnostics!

Linear and nonlinear models

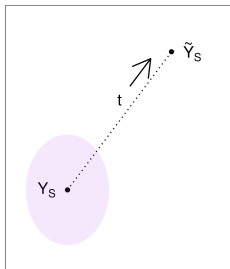
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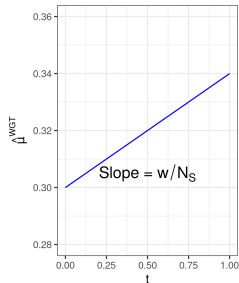
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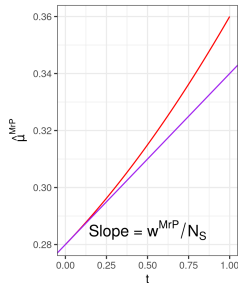
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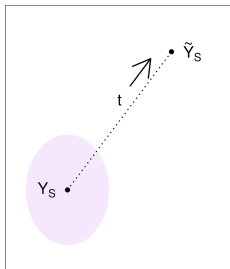
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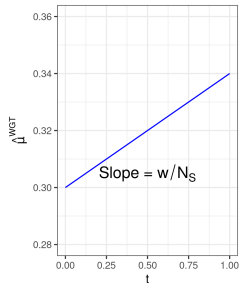
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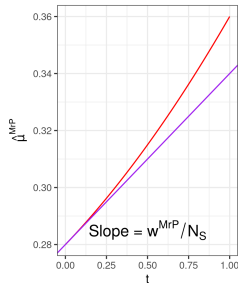
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$$w_i^{\text{MrP}} := N_S \frac{\partial \hat{\mu}^{\text{MrP}}(Y_S)}{\partial y_i} = N_S \text{Cov}_{\mathcal{P}(\theta|Y_S)} \left(\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top \mathbf{x}_j), \frac{\partial}{\partial y_i} \log \mathcal{P}(y_i | \mathbf{x}_i, \theta) \right)$$

Can estimate w_i^{MrP} without re-running MCMC [Gus96; GBJ18].

Same-Sex Marriage Survey Weights

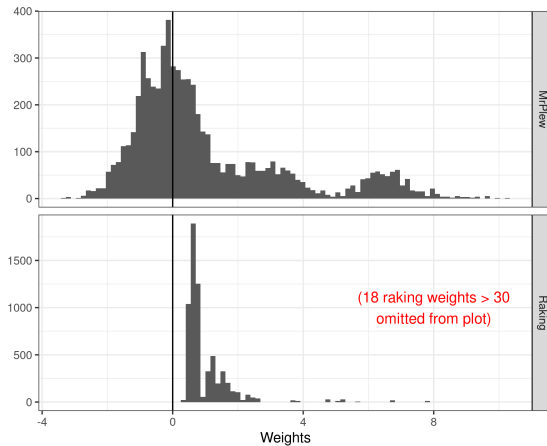


Figure 1: Comparison of weights for the Same-Sex Marriage analysis

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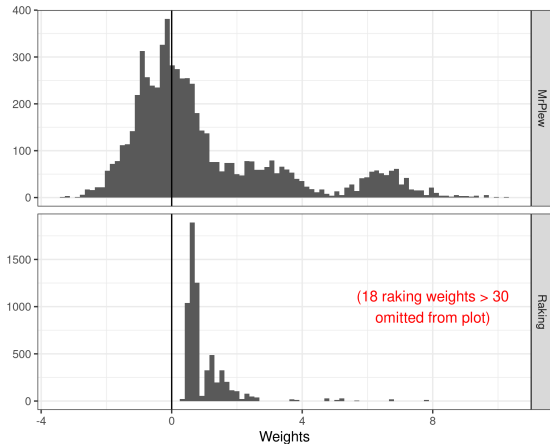


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What does the difference between these weights mean in terms of

- Frequentist sampling variance (standard errors)?
- Subgroup contributions?
- Covariate balance / external validity?

Same-Sex Marriage frequentist variance

Theorem [Gio+26] (informal):

You can use the weights for frequentist variance as if MrP were linear if (and only if) y_i is a sufficient statistic of the model.

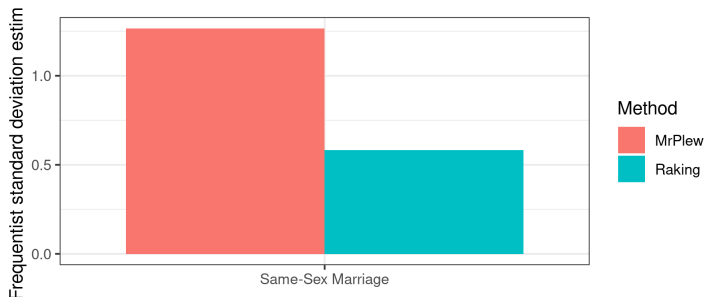
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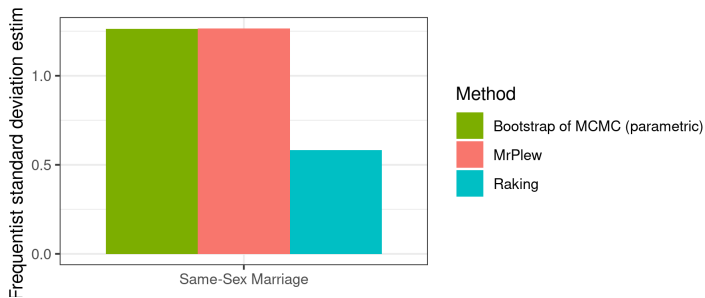


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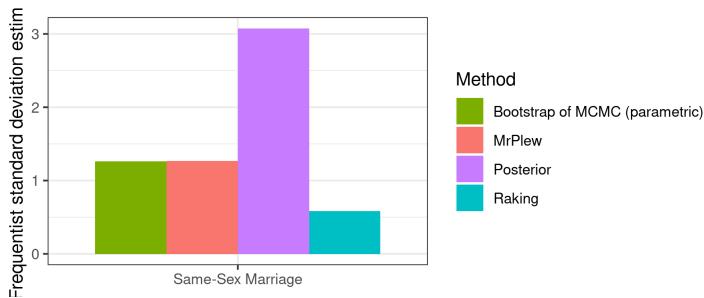


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Note that frequentist and posterior variance can be very different in misspecified models with many random effects!

Same-Sex Marriage subgroup contribution

Note that both methods use *national data* to infer public opinion in *California*.

TODO: put in plot of CA with question mark in Texas.

Question:

How do responses in Texas affect our estimate of public opinion in CA?

More specific question:

Suppose we *increased* the responses y_i in Texas.

How would our estimate of public opinion in CA change?

Theorem: [Gio+26] (informal)

Take $\tilde{y}_i = y_i + \delta f(\mathbf{x}_i)$. We state conditions under which

$$\left| \hat{\mu}^{\text{MrP}}(\tilde{Y}_S) - \hat{\mu}^{\text{MrP}}(Y_S) - \delta \sum_{i=1}^{N_S} w_i^{\text{MrP}} f(\mathbf{x}_i) \right| = \text{Small}$$

Uniformity helps justify searching over a large class of perturbations f .

² $\mathcal{F}$ can be any Donsker class of measurable functions with uniformly bounded $\mathbb{E} [\|\mathbf{x}\| f(\mathbf{x})] \leq \frac{2}{2}$.

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Suppose we *increased* the responses y_i in Texas.

How would our estimate of public opinion in CA change?

Theorem: [Gio+26] (informal)

Take $\tilde{y}_i = y_i + \delta f(\mathbf{x}_i)$. We state conditions under which

$$\left| \hat{\mu}^{\text{MrP}}(\tilde{Y}_S) - \hat{\mu}^{\text{MrP}}(Y_S) - \delta \sum_{i=1}^{N_S} w_i^{\text{MrP}} f(\mathbf{x}_i) \right| = O(\delta^2) \text{ as } N \rightarrow \infty$$

Uniformity helps justify searching over a large class of perturbations f .

² $\mathcal{F}$ can be any Donsker class of measurable functions with uniformly bounded $\mathbb{E} [\|\mathbf{x}\| f(\mathbf{x})] \leq \frac{2}{2}$.

Same-Sex Marriage subgroup contribution

Note that both methods use *national data* to infer public opinion *in California*.

TODO: put in plot of CA with question mark in Texas.

Question:

How do responses in Texas affect our estimate of public opinion in CA?

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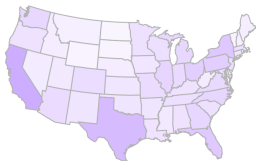
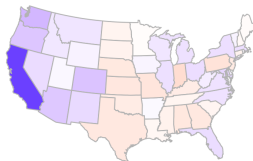
...for a very broad class of $\mathcal{F}$.²

Uniformity helps justify searching over a large class of perturbations f .

² $\mathcal{F}$ can be any Donsker class of measurable functions with uniformly bounded $\mathbb{E} [\|\mathbf{x}\| f(\mathbf{x})] \leq \frac{2}{2}$.

Same-Sex Marriage subgroup contribution

Summarize state contribution: $\sum_{n \text{ in state}} w_n$

Raking weight / N_s 
$$\text{MrPlew weight} / N_s$$


A vertical color bar legend for the 'Value' variable. The scale ranges from -1.0 at the bottom (red) to 1.0 at the top (blue), with intermediate ticks at -0.5, 0.0, and 0.5.

Figure 2: Subgroup contribution for Same-Sex Marriage in California

Same-Sex Marriage subgroup contribution

Summarize state contribution: $\sum_{n \text{ in state}} w_n$

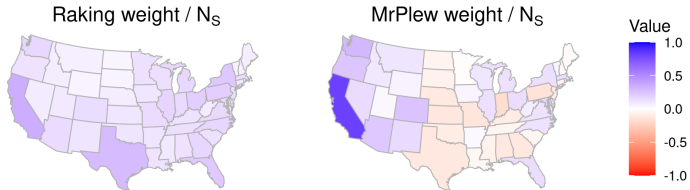


Figure 2: Subgroup contribution for Same-Sex Marriage in California

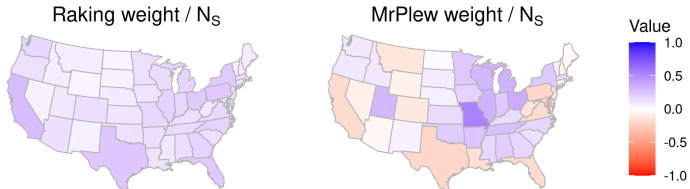


Figure 3: Subgroup contribution for Same-Sex Marriage in Missouri

By defining MrP locally equivlanent weights, we can rigorously provide a novel class of model checking metrics for non-linear MrP estimators.

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⚠ Stop! That is a misleading conclusion!

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⚠ Stop! That is a misleading conclusion!

MrP is modeling the response, not the sampling bias.

CW-like checks are not diagnostics for the response model.

A correctly specified response model can do badly on CW checks, and vice-versa! (See paper.)

This work is best thought of as *model interrogation* for MrP.

Conclusion (let's try that again)

By defining *MrP locally equivalent weights*, we can rigorously provide analogues of many CW diagnostics for non-linear MrP estimators for model interrogation.

(Similar ideas can to lead to actual checks of MrP model misspecification. Coming soon...)

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By defining *MrP locally equivalent weights*, we can rigorously provide analogues of many CW diagnostics for non-linear MrP estimators for model interrogation.

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R package available at

<https://github.com/rgiordan/mrplew>

Preprint at

https://rgiordan.github.io/assets/mrplew_paper.pdf

Extra slides

- [BFH21] Eli B., Avi F., and Erin H. *Multilevel calibration weighting for survey data*. 2021. arXiv: 2102.09052 [stat.ME].
- [Bür17] Paul-Christian Bürkner. “brms: An R Package for Bayesian Multilevel Models Using Stan”. In: *Journal of Statistical Software* 80.1 (2017), pp. 1–28. DOI: 10.18637/jss.v080.i01.
- [CZ23] A. Chattopadhyay and J. Zubizarreta. “On the implied weights of linear regression for causal inference”. In: *Biometrika* 110.3 (2023), pp. 615–629.
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- [Gel07a] A. Gelman. “Rejoinder: Struggles with survey weighting and regression modelling”. In: *Statistical Science* 22.2 (2007), pp. 184–188.
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- [Gus96] P. Gustafson. “Local sensitivity of posterior expectations”. In: *The Annals of Statistics* 24.1 (1996), pp. 174–195.
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Same-Sex Marriage Survey Weights

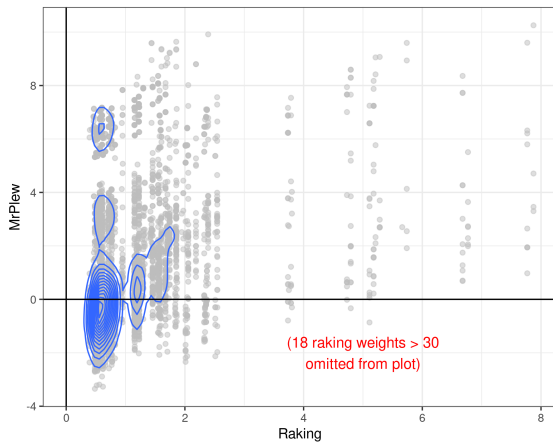


Figure 4: Comparison of weights for the Same-Sex Marriage analysis

Same-Sex Marriage covariate balance

Covariate balance / external consistency: $\frac{1}{N_T} \sum_{j=1}^{N_T} f(\mathbf{x}_j) \stackrel{\text{check}}{=} \frac{1}{N_S} \sum_{i=1}^{N_S} w_i f(\mathbf{x}_i)$

